

Ernest Choi

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Experience

Software Development Intern, MCG3D – Remote January 2026 – March 2026

- Built production web applications in React and TypeScript, keeping the component architecture scalable and the interface fast and user-centric.
- Integrated frontend features with Python Flask backend services over secure RESTful APIs, including AI-enabled workflows built on OpenAI APIs.
- Deployed and tested applications in Docker containers and wrote technical documentation, working within a remote, cross-functional team.

Electrical Team Member, Metropolitan Aerospace Rocket Society (MARS) – Toronto, ON September 2024 – Present

- Designed, fabricated, and integrated PCB circuits in Altium and KiCad for a custom STM32-based avionics flight computer with GPS telemetry, sensor integration, and pyrotechnic ignition.
- Improved power efficiency by 23% through circuit configuration, component selection, and power distribution across regulated voltage rails.
- Performed board bring-up and debugging with oscilloscopes, logic analyzers, and electrical measurements, validating hardware against strict performance and safety standards for flight.
- Developed embedded firmware for the microcontroller-based avionics in C/C++ and Python, automating test workflows with scripting.

Machine Learning Intern, Goopter Commerce Solutions – Remote September 2025 – December 2025

- Engineered a low-latency pipeline for quantitative trading, streaming high-frequency Kafka market data into asynchronous Python services on FastAPI, Pandas, NumPy, and scikit-learn.
- Built and trained trading-signal models in TensorFlow and PyTorch Lightning, operating on live market data streams.
- Deployed analytics dashboards in Tableau and Power BI for company use, surfacing real-time predictive insights from those models.

Technical Development Intern, City of Markham – Markham, Ontario July 2022 – August 2024

- Documented website workflows and operating processes for municipal web operations, helping cut website downtime by 41%.
- Tracked project milestones and reported progress to management in Microsoft Suite, reducing coordination delays by 14%.
- Validated and tested system changes against quality standards and functional requirements, reducing reported errors by 20%.

Projects

Particle Fluid Simulation | C++, OpenGL github.com/ErChoi-com/Particle-Fluid-Simulation

- Developed a real-time fluid simulation in C++ and OpenGL, writing GLSL shaders for particle rendering, physics-driven interaction, and GPU-accelerated parallel computation.
- Managed windowing, input, and render contexts with GLFW and GLUT, and trimmed graphics-pipeline memory allocation by 12% for smoother frames and better GPU utilization.
- Implemented a raycasting system that uses projection math to simulate depth and perspective.

Web Messaging App | JavaScript, Node.js, SQL ernestchoi.com/terminalcards

- Built a fullstack collaborative programming platform in JavaScript and Node.js, using WebRTC for real-time peer-to-peer messaging, file sharing, and room management.
- Implemented machine-learning autocomplete and in-browser Python code execution, alongside secure credential handling, key generation, and error recovery.
- Hosted the backend on a Raspberry Pi running Ubuntu, exposing a layered RESTful API backed by a secured PostgreSQL database.

Job-Monitoring & Resume-Generation System | Python, Playwright, LLM APIs, SQLite
github.com/ErChoi-com/DiscordBot

- Scraped and deduplicated 206,775 job postings (120,989 unique URLs) in one month across five ATS providers, using each source's REST API and escalating to Playwright only where required.
- Isolated failing sources behind per-source circuit breakers and a fair scheduler with concurrency caps, coordinated in asyncio so interactive commands preempt background scraping.
- Built an LLM pipeline returning structured JSON that Python renders deterministically into LaTeX, with provider fallback, SQLite storage, and a 500+ test pytest suite.

Skills

Languages: JavaScript/Node.js, Java, Python, C++, C, SQL, HTML, CSS, GLSL, VHDL, Verilog, MATLAB

Frameworks/Libraries: OpenGL, WebRTC, JavaFX, React, TypeScript, Flask, FastAPI, TensorFlow, PyTorch
Lightning, scikit-learn, Pandas, NumPy, Playwright, pytest

Tools: Git, PostgreSQL, SQLite, Ubuntu Linux, Docker, Kafka, Tableau, Power BI, Microsoft Suite

Platforms: GitHub, GitLab, Raspberry Pi, Arduino, FPGA

Relevant Courses: Object Oriented Eng Analysis and Design, Data Structures and Algorithms, Operating Systems, Computer Networks, Computer Organization and Architecture, Software Testing and Quality Assurance, Distributed Systems and Cloud Computing, Embedded Systems Design, Digital Systems, Digital Systems Engineering, Microprocessor Systems, Electronic Circuits I, Electronic Circuits II, Electric Networks, Signals and Systems, Control Systems, Discrete Mathematics, Probability and Statistics, Linear Algebra, Differential Equations

Education

Toronto Metropolitan University – Bachelor of Engineering in Computer Engineering

Expected 2028